

DINESH SHRESTHA

GIS Developer | Geospatial Automation | Enterprise GIS | Remote Sensing

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U.S. Permanent Resident - authorized to work without sponsorship | Previously held USGS Public Trust

PROFESSIONAL SUMMARY

GIS Developer and remote sensing professional with 8+ years of experience delivering geospatial solutions for utility, infrastructure, environmental, and federal programs. Combines enterprise GIS, spatial data engineering, automation, web GIS, and satellite-image analysis to turn operational and scientific requirements into reliable workflows, applications, and decision-ready data. Proven record improving data quality and reducing production time through reusable, documented solutions.

SELECTED CAREER IMPACT

90-95% Workflow Efficiency Reduced multi-hour mapping and editing workflows to minutes or seconds.	50+ Geospatial Data Layers Managed utility, environmental, infrastructure, survey, and scientific data.	1,200+ High-Resolution Maps Produced complex infrastructure and environmental cartographic deliverables.	10+ / 5+ Federal Standard Public Datasets / Publications Published 10+ USGS datasets and 5+ peer-reviewed publications.
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TECHNICAL SKILLS

GIS & Enterprise Platforms: ArcGIS Pro, ArcGIS Enterprise, Portal for ArcGIS, ArcGIS Server, ArcGIS Online, Experience Builder, Dashboards, Field Maps, Survey123, QGIS

Development & Automation: Python, ArcPy, SQL, pandas, GeoPandas, R, JavaScript, Arcade, JSON, HTML/CSS, ModelBuilder, GDAL, Jupyter, Git/GitLab

Spatial Data & Integration: Enterprise Geodatabases/SDE, PostgreSQL/PostGIS, Oracle Spatial, SQL Server, MySQL, spatial ETL, schema design, ArcGIS REST API, ArcGIS API for Python, map/feature services, data conversion and QA/QC

Spatial Analysis & Remote Sensing: Raster/vector analysis, image classification, change detection, vegetation and phenology analysis, viewshed analysis, spatial modeling, coordinate systems, raster masking, accuracy assessment

Domain Experience: Utility and transmission GIS, infrastructure mapping, environmental GIS, vegetation management, federal geospatial programs, survey/ROW data, cartography, field-data integration, Power BI reporting

PROFESSIONAL EXPERIENCE

GIS Specialist II | Eocene Environmental Group, Inc.

Mar 2023 - May 2026

- Developed reusable ArcGIS Pro and Python/ArcPy tools for utility and infrastructure workflows, including editing, feature updates, projection, archiving, Excel integration, QA/QC, and deliverable preparation.
- Automated a repetitive GIS editing process from approximately 7-10 minutes per feature to about 20 seconds by translating business rules into a standardized geoprocessing workflow.
- Automated Plan of Development map-series production for a ~300-mile high-voltage transmission project, generating 700-1,200-page mapbooks and reducing production time by approximately 90% - from 5-7 hours to about 30 minutes.
- Managed and reviewed access roads, ROWs, crossings, parcels, survey coverage, disturbance features, structures, attributes, geometry, projections, and deliverable completeness for utility and transmission projects.
- Built repeatable GIS workflows for vegetation management, wildfire mitigation, environmental review, transmission mapping, field-data integration, ArcGIS Online/Enterprise content, and Power BI reporting.
- Evaluated raster classification and agricultural-mask refinement for tree-canopy and land-cover analysis, including resampling, masking, edge effects, accuracy assessment, and spatial statistics.
- Troubleshoot ArcGIS Pro, Python environments, geoprocessing tools, schemas, and data sources while coordinating requirements and deliverables with BLM, USFS, Bureau of Reclamation, DOT, state agencies, clients, and multidisciplinary teams.

Remote Sensing Scientist II / GIS Analyst II | KBR Wyle Inc. - USGS EROS Center

Jul 2019 - Mar 2023

- Applied machine-learning methods, including XGBoost, with HLS imagery, vegetation indices, and environmental predictors to estimate fractional cover of multiple exotic annual grass species.
- Designed remote-sensing workflows for classification, resampling, masking, accuracy assessment, spatial statistics, and multi-sensor analysis across Landsat, Sentinel, HLS, MODIS, and VIIRS products.
- Configured and supported ArcGIS Enterprise/Portal, ArcGIS Server, ArcGIS Online, hosted layers, web maps, services, user access, and enterprise geodatabases for federal geospatial programs.
- Developed GIS applications, dashboards, map layouts, and user-facing spatial products using ArcGIS Enterprise/Online, Experience Builder, Arcade, Python, SQL, and Esri web technologies.

- Managed enterprise geodatabases/SDE, project repositories, schemas, metadata, and service-publishing workflows to improve data organization, traceability, retrieval, and usability.
- Built Python/ArcPy and SQL workflows for data ingestion, transformation, schema standardization, QA/QC, spatial analysis, reporting, and web/desktop GIS applications.
- Produced publication-ready datasets, maps, metadata, dashboards, technical documentation, and federal releases through USGS ScienceBase and related workflows.

Remote Sensing Scientist I / GIS Analyst I | KBR Wyle Inc. - USGS EROS Center

Jul 2017 - Jul 2019

- Performed image preprocessing, classification, change detection, raster/vector analysis, accuracy assessment, and large-volume batch processing using Python, R, ENVI/IDL, GDAL, Google Earth Engine, and ArcGIS.
- Developed CONUS-scale vegetation phenology and irrigated-agriculture products using MODIS/eMODIS and other time-series satellite data, including MirAD-US and seasonal phenology metrics.
- Developed and supported web, desktop, and mobile GIS applications using ArcGIS Pro/Desktop, ArcGIS Online, ArcGIS Server, ModelBuilder, Python/ArcPy, SQL, JavaScript, JSON, HTML/CSS, and shell scripting.
- Created, maintained, and quality-controlled enterprise geospatial datasets and services using Oracle Spatial, enterprise geodatabases, ArcGIS Server, and federal data-management workflows.
- Managed metadata, Data Management Plans, IPDS, QA/QC, data lineage, ScienceBase/USGS publication, and service integration while collaborating with scientists, developers, and federal partners.
- Worked with GIS specialists, scientists, developers, analysts, and federal partners to define requirements, solve spatial-data problems, and deliver reusable applications and datasets.

GIS Technician | Mid Dakota Vegetation Management

May 2017 - Jul 2017

- Produced GIS maps, vegetation-management work plans, reports, and Power BI visualizations using ArcMap, ArcCatalog, ModelBuilder, and field/tabular data.
- Maintained spatial datasets and GIS databases with Python/ArcPy and SQL, including cleanup, validation, automation, and repeatable data-management workflows.
- Integrated invasive-species field survey data into Power BI and developed dashboards and maps that helped field crews prioritize treatment areas and plan targeted spraying schedules.

GIS Teaching Assistant | Department of Geography, South Dakota State University

Aug 2015 - May 2017

- Facilitated GIS and geography labs for 500+ students covering spatial databases, raster/vector modeling, data integration, physical geography, and weather systems.
- Delivered ArcGIS demonstrations, software troubleshooting, office hours, and one-on-one technical support to help students apply GIS methods to practical problems.
- Collaborated with faculty, researchers, and students across disciplines to improve geospatial teaching methods and communicate analytical workflows clearly.
- Supported course delivery through organized lab preparation, student assistance, and clear technical communication across multiple GIS-focused classes.

SELECTED PROJECTS

- **MidAmerican Energy (Eocene):** Supported vegetation and wildfire-mitigation GIS through data management, QA/QC, Python/ArcPy automation, mapping, and Power BI tracking of circuits, work orders, schedules, and field operations.
- **Boardman to Hemingway (B2H) Transmission Line (Eocene):** Supported GIS analysis, data management, mapping, and automation for a ~300-mile transmission corridor and associated infrastructure, access, environmental, and field datasets.
- **SunZia Transmission Line (Eocene):** Performed schema standardization, QA/QC, spatial analysis, survey-data management, and custom geoprocessing for large-scale transmission infrastructure.
- **Southern Star Pipeline (Eocene):** Applied supervised classification and agricultural-land masking to improve tree-canopy and land-cover estimates along pipeline corridors.
- **Exotic Annual Grass Fractional Cover (USGS):** Applied XGBoost and remote-sensing predictors to estimate multi-year fractional cover of exotic annual grass species.
- **National Agriculture & Phenology Products (USGS):** Supported CONUS-scale irrigated-agriculture and vegetation-phenology products using MODIS/eMODIS imagery, geospatial processing, QA/QC, and federal publication workflows.

EDUCATION

M.S. Geography - GIS Specialization | South Dakota State University

Certificate in Data Science | South Dakota State University

Certificate in Machine Learning | 12-week professional course, KBR Wyle Inc.

B.E. Geomatics Engineering | Kathmandu University, Nepal